"local" Version of Thm 4.

Assume that $\mathcal{L}$ is a local operator on $C_c^\infty(\mathbb{R}^d)$ satisfying positive maximal principle, then

$$\mathcal{L}\varphi(x) = \frac{1}{2}a_{ij}(x)\,\partial_{ij}\varphi(x) + b_i(x)\,\partial_i\varphi(x) - c(x)\,\varphi(x).$$

where $a : \mathbb{R}^d \to \mathbb{R}^{d\times d}$, $a(x)$ is a nonnegative definite symmetric matrix;

$b : \mathbb{R}^d \to \mathbb{R}^d$;

$c : \mathbb{R}^d \to [0,\infty)$.

Pf: • We first show that if $\varphi(0) = \nabla\varphi(0) = \nabla^2\varphi(0) = 0$, then $L\varphi(0) = 0$. Let $\eta(x) = \eta(|x|)$. $\eta : \mathbb{R}_+ \to \mathbb{R}_+$ is smooth and decreasing, and $\eta(r) = 1$ if $r\in[0,1]$, $\eta(r) = 0$ if $r>2$.

Define $\varphi_R(x) := \varphi(x)\cdot\eta_R(x)$. Then for each $R\in(0,1)$.

$$|\varphi_R(x)| \le C|x|^3\,|\eta_R(x)| \le CR|x|^2\eta_R(x) \le Rf(x)$$

where $f(x) = C|x|^2\eta(x)$.

Thus $\quad RLf(0) \le L\varphi_R(0) = L\varphi(0)$.

$$L\varphi(0) \le L\varphi_R(0) \le -RLf(0).$$

Let $R\to 0$, one sees $L\varphi(0) = 0$.

• Now assume $\varphi\in C_c^\infty$. Let $f_0(x) = \eta(x)$, $f_i(x) = \partial_i\varphi(0)\,x_i\,\eta(x)$.

$f_{ij}(x) = \partial_{ij}\varphi(0)\,x_ix_j\cdot\eta(x)$. Let $a^0_{ij} = Lf_{ij}(0)$

$b^0_i = Lf_i(0)$, $\quad c^0 = -Lf_0(0) = -L\eta(0) \geq 0$.

Noting that $\phi = \varphi - \frac{1}{2}\sum_{i,j} f_{ij} - \sum_i f_i - f_0 \in C_c^\infty$,

$\phi(0) = \nabla\phi(0) = \nabla^2\phi(0) = 0$, by step 1, we get

$L\phi(0) = 0$, i.e.

$$L\varphi(0) = \frac{1}{2} a^0_{ij}\,\partial_{ij}\varphi(0) + b^0_i\,\partial_i\varphi(0) - c^0\varphi(0).$$

$\square$.